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import json
import os

FILE_NAME = "students.json"

def load_data():
    if os.path.exists(FILE_NAME):
        try:
            with open(FILE_NAME, "r") as f:
                return json.load(f)
        except json.JSONDecodeError:
            print("Warning: data file was
corrupted. Starting with empty records.")
            return {}
    return {}

def save_data(data):
    with open(FILE_NAME, "w") as f:
        json.dump(data, f, indent=4)

def get_valid_cgpa():
    while True:
        cgpa = input("Enter CGPA (0-10): ")
        try:
            cgpa_val = float(cgpa)
            if 0 <= cgpa_val <= 10:
                return cgpa_val
            print("CGPA must be between 0
and 10.")
        except ValueError:
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        print("Please enter a valid
number.")

def add_student(data):
    roll = input("Enter Roll Number: ")
    if not roll.isdigit():
        print("Roll number must be
numeric.")
    return
    if roll in data:
        print("Student with this roll
number already exists!")
    return
    name = input("Enter Name: ")
    branch = input("Enter Branch: ")
    cgpa = get_valid_cgpa()
    data[roll] = {"name": name, "branch":
branch, "cgpa": cgpa}
    save_data(data)
    print("Student added successfully!")

def view_students(data):
    if not data:
        print("No records found.")
        return
    for roll, info in data.items():
        print(f"Roll: {roll} | Name:
{info['name']} | Branch: {info['branch']} |
CGPA: {info['cgpa']}")

def view_sorted_by_cgpa(data):
    if not data:
        print("No records found.")
        return
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        sorted_students = sorted(data.items(),
key=lambda item: item[1]['cgpa'],
reverse=True)
        for roll, info in sorted_students:
            print(f"Roll: {roll} | Name:
{info['name']} | Branch: {info['branch']} |
CGPA: {info['cgpa']}")

def search_student(data):
    print("Search by: 1. Roll Number 2.
Name 3. Branch")
    option = input("Choose option: ")
    found = False
    if option == '1':
        roll = input("Enter Roll Number: ")
        if roll in data:
            info = data[roll]
            print(f"Found: {info['name']} |
{info['branch']} | CGPA: {info['cgpa']}")
            found = True
        elif option == '2':
            name = input("Enter Name (or part
of it): ").lower()
            for roll, info in data.items():
                if name in
info['name'].lower():
                    print(f"Roll: {roll} |
Name: {info['name']} | Branch:
{info['branch']} | CGPA: {info['cgpa']}")
                    found = True
        elif option == '3':
            branch = input("Enter Branch:
").lower()
            for roll, info in data.items():

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        if branch in
info['branch'].lower():
            print(f"Roll: {roll} |
Name: {info['name']} | Branch:
{info['branch']} | CGPA: {info['cgpa']}")
            found = True
        else:
            print("Invalid option.")
            return
    if not found:
        print("No matching student found.")

def update_student(data):
    roll = input("Enter Roll Number to
update: ")
    if roll not in data:
        print("Student not found.")
        return
    data[roll]['name'] = input("Enter new
Name: ")
    data[roll]['branch'] = input("Enter new
Branch: ")
    data[roll]['cgpa'] = get_valid_cgpa()
    save_data(data)
    print("Updated successfully!")

def delete_student(data):
    roll = input("Enter Roll Number to
delete: ")
    if roll in data:
        del data[roll]
        save_data(data)
        print("Deleted successfully!")
    else:

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        print("Student not found.")

def main():
    data = load_data()
    while True:
        print("\n1. Add 2. View 3.
Search 4. Update 5. Delete 6. View
Sorted by CGPA 7. Exit")
        choice = input("Enter choice: ")
        if choice == '1':
            add_student(data)
        elif choice == '2':
            view_students(data)
        elif choice == '3':
            search_student(data)
        elif choice == '4':
            update_student(data)
        elif choice == '5':
            delete_student(data)
        elif choice == '6':
            view_sorted_by_cgpa(data)
        elif choice == '7':
            break
        else:
            print("Invalid choice.")

if __name__ == "__main__":
    main()
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